

Data Dictionary — Inventory_Finance_StarSchema_5000Rows.xlsx

Schema type: Star schema — 4 dimension tables + 1 fact table **Fact table grain:** One row per inventory/batch record for a given Product × Location × Customer × Date

Relationships: Fact_Inventory connects to each dimension via its surrogate key (DateKey, ProductKey, LocationKey, CustomerKey)

1. Dim_Product (200 rows)

Grain: One row per material/product (SKU)

Column	Type	Description	Values / Range
ProductKey	Integer (PK)	Surrogate key uniquely identifying the product	1-200
Mat Code	Text	Business/material code	MAT0001-MAT0200
Material Description	Text	Product name	e.g. "Coating Product 1"
CAT	Text	Product category	Coating, Chemical, Paint, Adhesive
Mat Gp	Text	Material group code	WD, IND, EXT, INT
Material Group Desc	Text	Material group description (maps to Mat Gp)	Wood, Industrial, Exterior, Interior
Brand	Text	Product brand	Alpha, Prime, Nova, Ultra
ShelfLifeDays	Integer	Shelf life of the product in days	365, 540, 730
ABC Class	Text	ABC inventory classification (value-based ranking)	A, B, C

2. Dim_Date (730 rows)

Grain: One row per calendar day, Jan 1, 2025 - Dec 31, 2026 (2 full years)

Column	Type	Description	Values / Range
DateKey	Integer (PK)	Surrogate key for the date	1-730
Date	Date	Calendar date	2025-01-01 to 2026-12-31
Year	Integer	Calendar year	2025, 2026
Quarter	Integer	Calendar quarter	1-4
Month	Integer	Calendar month number	1-12
MonthName	Text	Calendar month name	January-December

3. Dim_Location (20 rows)

Grain: One row per plant/warehouse/sales-office combination

Column	Type	Description	Values / Range
LocationKey	Integer (PK)	Surrogate key for the location	1-20
Plant	Text	Plant name	Plant 1-Plant 20
Warehouse	Text	Warehouse code (1:1 with Plant)	WH-01-WH-20
Sales Office Desc	Text	Sales office code (1:1 with Plant)	SO-01-SO-20
Zone	Text	Geographic zone	East, West, North, South
Country	Text	Country	India (single value across all rows)

4. Dim_Customer (500 rows)

Grain: One row per customer

Column	Type	Description	Values / Range
CustomerKey	Integer (PK)	Surrogate key for the customer	1-500
Customer Code	Text	Business customer code	CUST0001-CUST0500
Customer Name	Text	Customer name	e.g. "Customer 1"
Customer Type	Text	Customer channel type	Dealer, Retailer, Distributor

5. Fact_Inventory (5,000 rows)

Grain: One row per inventory batch transaction, linked to a specific Product, Location, Customer, and Date

Column	Type	Description	Values / Range
InventoryID	Integer (PK)	Unique identifier for the fact row	1-5,000
DateKey	Integer (FK → Dim_Date)	Links to the transaction date	1-730
ProductKey	Integer (FK → Dim_Product)	Links to the product	1-200
LocationKey	Integer (FK → Dim_Location)	Links to the plant/warehouse	1-20
CustomerKey	Integer (FK → Dim_Customer)	Links to the customer	1-500
SEGMENT	Text	Business segment for the transaction	Projects, Retail, Industrial
Quantity	Integer	Units sold/transacted	5-500
Sales Value INR	Decimal	Sales revenue in INR	₹187.82-₹584,897.55
TOTAL Gross Margin	Decimal	Gross margin (INR) earned on the transaction	₹40.51-₹188,749.92
InventoryQty	Integer	On-hand inventory quantity at time of record	0-2,500
UnitCost	Decimal	Cost per unit (INR)	₹20.04-₹799.74
InventoryValue	Decimal	Total inventory value (InventoryQty × UnitCost, approx.)	₹0.00-₹1,973,985.09
BatchNumber	Text (unique)	Batch/lot identifier	B100002-B105001
StockInDate	Date	Date stock was received into inventory	2024-03-23 to 2026-12-08
ManufacturingDate	Date	Date the batch was manufactured	2024-03-13 to 2026-11-29
ExpiryDate	Date	Date the batch expires (Manufacturing Date + Shelf Life)	2025-03-13 to 2028-11-16

Column	Type	Description	Values / Range
LastSaleDate	Date	Most recent date the batch was sold	2024-11-05 to 2026-12-28
DaysInInventory	Integer	Number of days the batch has been in inventory	15-299
InventoryStatus	Text	Movement classification	Healthy, Slow Moving, Fast Moving
ReorderLevel	Integer	Inventory threshold that triggers reordering	100-500
SafetyStock	Integer	Minimum buffer stock to hold	50-150
LeadTimeDays	Integer	Supplier/replenishment lead time in days	5-30

Entity Relationships



All four dimension tables join to **Fact_Inventory** on a 1-to-many basis via their respective surrogate keys.